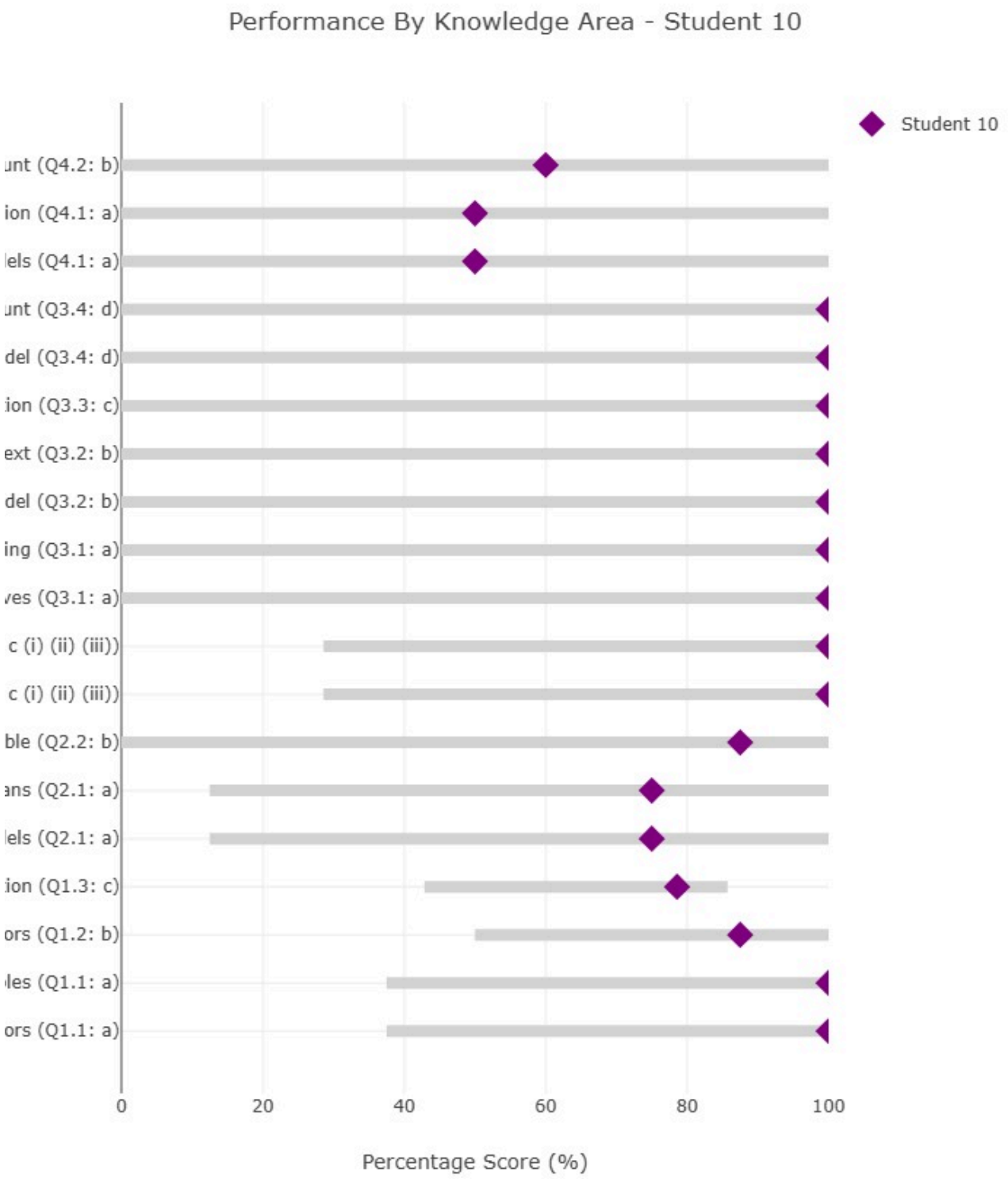
Student feedback for Student 10

The chart belwo shows your performance across each knowledge area assessed. The grey line shows the range of scores achieved by the rest of the cohort. The purple marker shows your own score within that range.



Dear student, you achieved **84/100**, which is a very strong result across a technically demanding

assessment. Your work shows confidence with regression modelling, classification, prediction, and model evaluation. You are clearly close to the level where a fully polished solution could reach 100%, so the main aim now is to remove the small gaps in interpretation and diagnostic checking.

In Question 1, you performed very well overall, scoring **8/8 for 1.1**, **7/8 for 1.2**, and **11/14 for 1.3**. Your answer to 1.1 showed a secure grasp of the structure of a linear model with factor variables, and you fitted the model appropriately. In 1.2, your interpretation of the slope was mostly accurate; to gain the final mark, be especially precise that the slope represents the expected change in body mass for a one millimetre increase in flipper length, holding the other predictors constant. For 1.3, your prediction method was largely correct, but the lost marks suggest that either the substitution into the fitted equation or the final numerical value needed closer checking.

Question 2 was also strong, with **6/8 for 2.1**, **7/8 for 2.2**, and an excellent **14/14 for 2.3**. Your full-mark logistic regression response was a particular strength: you clearly understood the form of the logit model and applied it correctly. In 2.1, the mixed-effects model was broadly handled well, but you should revisit how random effects alter predicted means for each island, especially when predictions are requested at the average flipper length. In 2.2, creating the binary Gentoo indicator was nearly correct; small coding or labelling details are often where marks are lost in otherwise good answers.

Your performance in Question 3 was outstanding, with **5/5 on all four parts**. This means you successfully handled the train-test split and ROC comparison in 3.1, calculated and interpreted the odds ratio in 3.2, filtered the dataset correctly in 3.3, and fitted the Poisson model in 3.4. This section shows very strong applied skill and good fluency moving between different modelling frameworks.

The main area for improvement is Question 4, where you scored **5/10 for 4.1** and **6/10 for 4.2**. In 4.1, the key issue is likely the depth of your explanation of overdispersion in count data models: make sure you can state what is being compared, why overdispersion matters, and what it implies for standard errors and inference. In 4.2, your confidence interval plot was partly successful, but you should work on ensuring the predictions, intervals, and male-student subset are all aligned correctly and clearly labelled.

These resources can help you build your understanding in the areas where you did not score as highly.

1. Mixed-effects models in R
2. Overdispersion in Poisson regression
3. Model predictions and confidence intervals in R

Overall, this is an impressive formative result. You have already demonstrated that you can handle most of the core statistical modelling skills at a high level. For the next test, focus on precision in interpretation, careful numerical checking, and clearer diagnostic reasoning. If you do that, a score much closer to **100%** is entirely realistic.